

Smarajit Paragranjan Mishra

Electronics and Communication Engineering (ECE) student

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Profile

I'm Smarajit, an Electronics and Communication Engineering (ECE) student with a strong interest in VLSI technologies. I am highly motivated, quick to learn, and passionate about applying my knowledge to real world challenges. I take pride in being a team player with a proactive attitude and a commitment to continuous growth. Outside academics, I enjoy playing badminton and watching mythological TV shows. I'm always eager to explore new technologies and enjoy working with others to find smart, practical solutions.

Work Experience

05/2024 – 06/2024

Baleswar

Intern

Defense Research and Development Organisation (DRDO)

- Worked in the Telecommand Unit (Flight Termination System), learning Ground Telecommand (GTC) and On-Board Telecommand (OBTC) system operations for missile range safety.
- Developed signal processing & computational tools in NI LabVIEW, including sine wave generation, amplitude modulation with FFT analysis, and mean/variance calculators.
- Gained exposure to RF communication links, safety protocols, and hardware software integration in missile testing environments.

05/2023 – 07/2023

Vani Vihar,

Bhubaneswar

Intern

Regional Telecommunication training center

Completed a 6-week online training on ADVANCE TELECOM, enhancing my skills in Global System for Mobile Communications (GSM), Digital Transformation, Optical Fiber, Communication, IPv4, Computer Networking, Data communication

Education

08/2025 – 05/2027

Bhubaneswar

M.TECH IN VLSI DESIGN AND EMBEDDED SYSTEMS

ODISHA UNIVERSITY OF TECHNOLOGY AND RESEARCH

Current CGPA: 9.44 / 10 (till 1st semester)

11/2021 – 05/2025

B.TECH IN ECE

SAMBALPUR UNIVERSITY INSTITUTE OF INFORMATION TECHNOLOGY

CGPA: 8.61 / 10

Skills

Programming Languages

Python, VHDL, Verilog

Soft Skills

Problem-Solving, Critical Thinking, Team Collaboration & Technical Communication

Analytical & Design Skills

CMOS Logic Design, Semiconductor Device Fundamentals, Technology Scaling

Electronics & Hardware

Circuit Analysis, Oscilloscope Operation, SMD Hand Soldering

Design & Simulation Tools

Silvaco TCAD (ATLAS), LTspice, Electric VLSI (Layout & Schematic), Xilinx ISE, Vivado

Projects

- 03/2026 – 04/2026 **Design and Implementation of High-Speed 8-Bit Multiplier Using Hybrid Adder-Based 5:3 Compressor**
- Designed and implemented a high-speed 8-bit multiplier using a hybrid adder-based 5:3 compressor architecture in Verilog HDL.
 - Optimized partial product reduction by integrating 5:3 compressors, Full Adders, and Half Adders, reducing logic depth and improving computational speed.
 - Achieved a post-implementation critical path delay of 8.845 ns with only 78 LUTs (0.15% FPGA resource utilization) on the Xilinx Zynq-7000 platform.
- 11/2025 – 12/2025 **Design and Analysis of Dual Material Gate TFET using Silvaco TCAD**
- Designed and simulated a Dual Material Control Gate Tunnel Field Effect Transistor (DMCG-TFET) using Silvaco ATLAS TCAD.
 - Implemented gate work-function engineering and introduced a metal strip in the oxide region near the source–channel junction to enhance electric field.
 - Achieved significant improvement in ON-state current from 2.52×10^{-6} A to 4.47×10^{-5} A while maintaining ultra-low OFF current ($\sim 10^{-17}$ A).
 - Improved ION/IOFF ratio to approximately 3.88×10^{12} , demonstrating better switching performance for low-power applications.
 - Analyzed device characteristics including transfer curves (IDS-VGS), energy band diagrams, and threshold voltage behavior using TCAD simulations.
- 10/2025 – 11/2025 **CMOS Priority Encoder using LTspice**
- Designed and simulated an 8-to-3 CMOS priority encoder in LTspice.
 - Evaluated power and timing behavior across 250 nm to 16 nm technology nodes.
 - Observed strong power scaling trends and improved performance at lower nodes.
- 01/2025 – 04/2025 **WALLACE TREE MULTIPLIER USING BRENT KUNG ADDER**
- The critical path delay was reduced from 18.25 ns in the Classical Multiplier to 12.38 ns in the Wallace Tree Multiplier, and further to 10.65 ns in the Wallace Tree with Brent-Kung Adder design, resulting in a 41.6% overall improvement in speed.
 - The total power consumption decreased from 0.354 W in the Classical Multiplier to 0.312 W in the Wallace Tree Multiplier, and further to 0.278 W in the Wallace Tree with Brent-Kung Adder design, achieving a 21.4% overall reduction in power usage.
 - The FPGA resource utilization (LUTs and slices) was reduced by 15% when moving from the Classical Multiplier to the Wallace Tree Multiplier, and by an additional 9% with Brent-Kung Adder integration.
- 11/2024 – 12/2024 **DESIGN OF 16-BIT RIPPLE CARRY AND BRENT-KUNG ADDERS USING HDL**
- The critical path delay was reduced from 10.98 ns in the Ripple Carry Adder to 9.43 ns in the Brent-Kung Adder, achieving a 14% improvement in speed.
 - The total power consumption decreased from 0.301 W to 0.266 W, representing an 11.6% reduction in power usage.
 - The FPGA resource utilization (LUTs and slices) was reduced by 8%, leading to more efficient hardware usage.
- 01/2023 – 05/2023 **FIRE & GAS DETECTOR SYSTEM WITH PROTECTION USING ARDUINO NANO**
- Designed and developed a safety system using Arduino Nano, integrating MQ-2 gas sensor and IR flame sensor for early detection of hazardous gas leaks and fire.
 - Implemented automatic control of buzzer, exhaust fan, and extinguisher activation upon detection, with real-time status displayed on a 16x2 LCD via I2C driver.
 - Programmed and interfaced hardware components using C/C++ on Arduino IDE, with serial communication for live monitoring and debugging.
 - Achieved a stable, reliable response with minimized false alarms, ensuring accurate hazard detection and prompt safety actions.

Language

Hindi
Fluent

English
Fluent

Odia
Native